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Received: 5 December 2025

Accepted: 13 July 2026

Published online: 29 July 2026

Cite this article as: Ying, X., Ninan, K., Oberste, J.-P. *et al.* An automated approach to improve clinical trial registration and to identify outcome changes on ClinicalTrials.gov. *npj Digit. Med.* (2026). <https://doi.org/10.1038/s41746-026-03045-y>

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An automated approach to improve clinical trial registration and identify outcome changes on ClinicalTrials.gov

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Word count: 4,609

Abstract

It is difficult to ensure clinical trial outcomes are defined completely in registrations, and to check outcome changes between registration and posting results. We developed a large language model (LLM)-based approach to evaluate outcome definitions and changes on ClinicalTrials.gov. Our LLM-based approach accurately identified incomplete outcomes in prospective trial registrations (sensitivity, 0.91 [95% confidence interval {CI}, 0.87–0.94]; positive predictive value [PPV], 0.98 [95% CI, 0.97–1.00]). Comparing prospective registrations with posted results, it correctly identified 96.1% of missing and 97.9% of added outcomes. It identified all outcomes with a priority change (e.g., from primary to secondary) and 99.2% without one. Sensitivity and PPV for identifying any outcome change were 0.97 (95% CI, 0.95–0.99) and 0.95 (95% CI, 0.89–0.98), respectively. Estimated costs per trial were \$0.13 for o3-mini, \$0.30 for GPT-4o, and \$1.80 for o1. The accuracy of our approach suggests it could be used to improve registration quality and to detect outcome changes at large scale and at low cost.

Introduction

Reporting results for planned outcomes is essential for clinical trials to be credible and useful. Laws and policies aim to reduce reporting bias by requiring that clinical trial investigators register randomized controlled trials (RCTs) in trial registries such as ClinicalTrials.gov.¹⁻⁶ Because trial protocols and statistical analysis plans (SAPs) are not always publicly available, trial registrations are often the only publicly accessible source of information on a trial's prespecified plans. Investigators are expected to define outcomes before conducting clinical trials, and they are expected to post results for planned outcomes at the end of their research, so that readers can assess whether planned outcomes are consistent with the outcomes for which results are reported.⁷

Changes to planned outcomes are common.⁸⁻¹¹ While some changes improve trial quality (e.g., in response to emerging evidence), changes made after data collection or analysis can introduce selective reporting bias, threaten the validity of findings, and mislead evidence users. Prior meta-research examining outcome changes has relied on manual review. Consequently, meta-research has been limited in scale, and it has often relied on expert judgement using implicit rules that would be difficult to explain or reproduce.¹²⁻²¹

An outcome is one of five components of an estimand. For studies of regulated products, ICH E9 (R1) emphasizes that an estimand should be specified in a protocol and statistical analysis plan, but it does not provide guidance for defining outcomes completely.^{22,23} Zarin et al. (2011) previously proposed a structured framework for defining outcomes,²⁴ which was adopted by ClinicalTrials.gov and later incorporated in guidance such as SPIRIT and CONSORT.²⁵⁻²⁸ As we have described elsewhere,²⁹ current best practice is to define an outcome completely by reporting the construct of interest, data collection instruments, method for assigning values to individuals, cutoff used to define the outcome (if applicable), whether the values in individuals are treated as continuous or categorical, and the time(s) at which data will be collected. Completely defined outcomes minimize ambiguity and limit opportunities for selective reporting (i.e., minimize risk of reporting bias).³⁰

There are several potential barriers to registering planned outcomes completely. Investigators and trial sponsors (e.g., universities, drug or device manufacturers) might not know what is required to define an outcome, and they might not understand how outcome definitions will affect the interpretation of their research. For example, incomplete outcome definitions in registers such as ClinicalTrials.gov can lead to concerns about *risk* of selective reporting bias even when investigators adhere to prespecified methods. Sponsors, especially non-industry sponsors, might lack resources to check the completeness of outcome definitions before submitting registrations.³¹ Staff at registries such as ClinicalTrials.gov review records manually to confirm whether all required information is present (Box 1),³² which can be time consuming and lead to multiple revisions and delayed posting.³³

Editors, peer reviewers, and evidence users find it difficult to assess whether planned outcomes match reported outcomes and results. Trials often assess multiple outcomes, which investigators might reorder after initial registration, so matching outcomes across registration versions can be challenging. Moreover, the elements that define each outcome in a registration are embedded within unstructured text. Text might change while trials are underway, but textual changes are not always important and might not indicate that outcome definitions have changed. Previous studies show that barriers to evaluating registrations during peer review include time, perceived usefulness, lack of awareness, and complexity.³⁴ For risk of bias assessments in systematic reviews, selective outcome reporting is considered most difficult to assess.^{35,36}

Large language models (LLMs), such as Generative Pre-trained Transformer (GPT) models, have demonstrated strong capabilities in understanding and comparing text. LLMs can also provide explanations for their decisions. Structured, automated outcome assessments using LLMs could help investigators, sponsors, and registries improve the completeness and consistency of outcome definitions before trials begin. Automated assessments could also enable faster identification of outcome changes and large-scale meta-research.

In this study, we developed and evaluated an automated approach to determine whether outcomes on ClinicalTrials.gov are defined completely and to identify changes between registration versions. We also explored whether model performance varied across trials subject to different regulatory requirements and levels of quality control.

Box 1. Fields for registering outcomes on ClinicalTrials.gov^{29,37}

Outcome Measures Module: Includes information about all outcomes. Outcomes are organized under the headings Primary, Secondary, and Other; each category may include multiple outcomes.

Outcome Measure Title: The exact measurement to be used (e.g., the name of a scale, physiological parameter, or questionnaire) and the metric and variable type used to aggregate data (e.g., mean change from baseline). It is inappropriate to enter only a broad outcome domain (e.g., “feasibility,” “safety,” “pain,” “depression”) in this field.

Outcome Measure Description:

Additional details about the measurement, metric, or variable type noted in the title. For example, if the measure is “participants with response,” the description should define the criteria for “response,” such as a specific cutoff. For unpublished questionnaires, the description may include the number and types of questions, the characteristics assessed, the directionality and interpretation of scores, and other relevant details. Investigators may also specify the planned statistical methods for comparing groups, which is related but not part of the outcome definition.

Outcome Measure Time Frame: The specific time point or time points at which the outcome will be assessed. Investigators should indicate the times for which data will be analyzed and reported. Times may include the start point, end point, and any intermediate points in between. All time points should be expressed from the participant’s perspective (e.g., use “from enrollment to end of treatment at 8 weeks,” not “until end of study”). Whenever applicable, a standard unit of time should be specified. For time-to-event outcomes, the Time Frame should describe the estimated period over which the event is assessed (e.g., “from the date of assignment until the date of first documented progression or death from any cause, whichever comes first, assessed up to 100 months”). If the outcome measures change from baseline, change between two time points, or change over time, the Time Frame should specify all relevant points, including the baseline if it is used to calculate change (e.g., “at baseline and at 6 months”). If multiple time points are entered, it is preferable to clarify whether they will be analyzed together in a single model (e.g., mixed-effects analysis) or to specify which time points are primary.

Edit Outcome Measures
[← Back to Outcome Measures](#)

Use this module to describe the primary, secondary, and other pre-specified outcome measures that will be collected and analyzed.

- All primary and secondary outcome measures that are listed in the study's protocol must be reported.
- At least one primary outcome measure must be entered.

* Required
 * § Required if Study Start Date is on or after January 18, 2017
 ⓘ Conditionally required

Outcome Measure Type

- ☒ Primary Outcome Measure
☐ Secondary Outcome Measure
☐ Other Pre-Specified Outcome

Primary Outcome Measure * ⓘ**Title * ⓘ**

253 characters left

Description ⓘ ⓘ

999 characters allowed

Time Frame * ⓘ

253 characters left

Save Changes[Clear Unsaved Changes](#)

Results

Characteristics of the validation trials

From more than 477,000 registered studies, we identified 4,574 trials meeting our eligibility criteria (Supplementary Figure 1). From these, we used stratified sampling to select 150 validation trials. Because of input and output token limits in GPT models (GPT-4o-2024-11-20: Max context 128,000 tokens; Max output tokens 16,384 tokens), we were unable to process one trial in the initial set of 150 validation trials, which included 45 outcomes. We replaced it with another randomly selected trial.

From 150 included trials, we identified 2,292 outcomes, including 1,032 in prospective registrations and 1,260 in latest registrations (Table 1). Human raters identified 1,658 matched outcomes (i.e., 829 pairs) and 634 unmatched outcomes in either of the two versions; 19.7% of prospectively registered outcomes were missing from the latest registration. Of the outcomes in the latest registrations ($n = 1,260$), 34.2% were added, 55.2% differed, and 10.6% were unchanged compared with prospective registrations. The most frequent type of change was *more complete* for specific measurement (12.8%), specific metric (13.9%), and variable type (66.3%), whereas *change in definition* was most common for cutoff (7.5%) and time point (34.2%) (Supplementary Table 1).

Human raters evaluated the completeness of outcome definitions in 75 trials, including 538 outcomes in prospective registrations and 607 outcomes in the latest registrations. We identified 12.8% and 74.5% of outcomes in the prospective and latest registrations as *defined completely*, respectively. Only 21.9% of prospective registrations defined the variable type (Supplementary Table 2).

Percentage agreement between human raters was 97.8% for matching, 95.0% for any change, and 80.2% to 96.2% for changes in specific elements (Supplementary Table 3).

Table 1. Characteristics of RCTs and outcomes in the validation set

	Total	Applicable clinical trials	NIH-funded trials	Other trials
Trials (N)	150	50	50	50

Phases				
Not Applicable*	105 (70.0%)	18 (36.0%)	47 (94.0%)	40 (80.0%)
Phase III	45 (30.0%)	32 (64.0%)	3 (6.0%)	10 (20.0%)
Registered on or after January 18, 2017	73 (48.7%)	28 (56.0%)	22 (44.0%)	23 (46.0%)
Intervention type [‡]				
Behavioral	44 (29.3%)	0 (0.0%)	30 (60.0%)	14 (28.0%)
Device	27 (18.0%)	19 (38.0%)	2 (4.0%)	6 (12.0%)
Drug	48 (32.0%)	26 (52.0%)	5 (10.0%)	17 (34.0%)
Other	31 (20.7%)	5 (10.0%)	13 (26.0%)	13 (26.0%)
Health condition				
Cancer	17 (11.3%)	6 (12.0%)	4 (8.0%)	7 (14.0%)
Cardiovascular & Metabolic Diseases	26 (17.3%)	8 (16.0%)	9 (18.0%)	9 (18.0%)
Mental & Behavioral Health	27 (18.0%)	5 (10.0%)	16 (32.0%)	6 (12.0%)
Neurological & Pain Disorders	15 (10.0%)	8 (16.0%)	3 (6.0%)	4 (8.0%)
Other	65 (43.3%)	23 (46.0%)	18 (36.0%)	24 (48.0%)
Number of registration versions at the time of our study				
Median (interquartile range)	11 (8 to 17)	13.4 (9 to 29)	10 (8 to 13)	9.5 (6 to 20)
Mean (standard deviation)	23.1 (69.2)	43.5 (117.1)	10.7 (3.7)	14.9 (13.6)
Outcomes in prospective registrations (N)	1032	401	339	292
Outcomes per trial, mean (SD)	6.9 (5.0)	8.0 (5.6)	6.8 (5.2)	5.8 (3.9)
Incomplete [†]	469 (87.2%)	201 (89.3%)	154 (86.5%)	114 (84.4%)
Missing	203 (19.7%)	107 (26.7%)	57 (16.8%)	39 (13.4%)
Change in definition or completeness	696 (67.4%)	259 (64.6%)	230 (67.8%)	207 (70.9%)
Outcomes in latest registrations (N)	1260	426	405	429
Number of outcomes, mean (SD)	8.4 (6.0)	8.5 (6.6)	8.1 (6.2)	8.6 (5.1)
Incomplete	155 (25.5%)	56 (27.6%)	61 (31.3%)	38 (18.2%)
Added	431 (34.2%)	132 (31.0%)	123 (30.4%)	176 (41.0%)
Change in definition or completeness	696 (55.2%)	259 (60.8%)	230 (56.8%)	207 (48.3%)

Data source: ClinicalTrials.gov.

*In ClinicalTrials.gov, *Not Applicable* is used to describe trials without FDA-defined phases, including trials of devices or behavioral interventions.

[‡] Our sample did not include RCTs of the following intervention types (as defined by ClinicalTrials.gov): genetic (e.g., stem cell), combination product (e.g., drug-device combination), and diagnostic test (e.g., imaging).

[†] Outcome definition completeness was assessed in 75 trials which include 25 ACTs, 25 NIH-funded trials, and 25 Other trials.

Tokens and cost

To complete all tasks—including defining outcomes, matching outcomes, and comparing outcomes—the average input length was 90,907 tokens, with an average output for GPT-4o of 7,329 tokens. Estimated costs were \$0.13, \$0.30, and \$1.80 per trial for o3 mini, GPT-4o, and o1, respectively. On average, GPT-4o completed the entire process (i.e., matching, defining, comparing) in 96 seconds per trial, and it took humans approximately 27 minutes per trial to evaluate GPT’s response.

Identifying incomplete outcome definitions

Sensitivity for identifying incomplete outcomes in prospective registrations was 0.91 (95% confidence interval [CI]: 0.87, 0.94) and specificity was 0.88 (95% CI: 0.79, 0.97) (Figure 1). In the latest registrations, sensitivity for identifying incomplete outcomes was lower (0.67, 95% CI: 0.54, 0.80) whereas specificity was higher (0.94, 95% CI: 0.91, 0.96) (Supplementary Figure 2).

Defining and explaining outcomes

Percent agreement between our approach and human raters exceeded 85% for every element (Figure 2). Patterns across elements were similar for prospective and latest registrations, with the highest agreement for cutoff (95.6% and 93.9%, respectively) and the lowest agreement for specific measurement (86.5% and 88.3%, respectively). Box 2 includes examples of automated explanations that describe what information was present and what information would be needed to define outcomes completely.

Box 2: Example of LLM output for defining outcomes using information from ClinicalTrials.gov registration NCT03259074

Outcome information extracted from ClinicalTrials.gov:

Title: No radiographic progression as measured by modified Stoke Ankylosing Spondylitis Spine Score (mSASSS)

Description: To demonstrate the proportion of subjects on secukinumab (combined 150 mg s.c. and 300 mg s.c.) with no radiographic progression as measured by mSASSS at Week 104 is superior to subjects on GP2017 (adalimumab biosimilar 40 mg s.c.)

Timeframe: 104 weeks

Specific measurement identified by a LLM (o1):

Definition “Modified Stoke Ankylosing Spondylitis Spine Score (mSASSS)”

Rationale: The outcome measures radiographic progression in ankylosing spondylitis patients using the modified Stoke Ankylosing Spondylitis Spine Score (mSASSS). Radiographic progression is a Subjective or Complex attribute because it involves disease progression, which cannot be directly observed or quantified objectively. It is assessed using a specialized tool, the mSASSS. Following the instructions for naming a specialized tool or questionnaire, we provide the full name and include the acronym.

Specific metric identified by a LLM (GPT-4o):

Definition: “Value at a time point”

Rationale: The outcome measures the proportion of subjects with no radiographic progression as measured by mSASSS at Week 104. This is a binary outcome indicating whether or not progression occurred, and it is assessed at a single time point without involving calculations across time points.

Cutoff identified by a LLM (o1):

Definition: “Not reported”

Rationale: The outcome is 'No radiographic progression as measured by mSASSS'. However, No specific cutoff is provided to define 'no radiographic progression'. Therefore, the 'Cutoffs' is 'Not reported'.

Variable type identified by a LLM (GPT-4o):

Definition: "Categorical"

Rationale: The outcome is described as the proportion of subjects with no radiographic progression, which is typically analyzed categorically. The text explicitly mentions 'proportion', confirming categorical aggregation.

Matching outcomes

Our automated approach correctly matched 99.0% (821/829) of outcomes, and it correctly identified 96.1% (195/203) and 97.9% (422/431) of missing and added outcomes, respectively.

Comparing outcome definitions and priorities

Our automated approach identified outcomes that changed with high sensitivity (0.97 [0.95, 0.99]) and PPV (0.95 [0.89, 0.98]) (Figure 3). It accurately identified the absence of changes for specific measurement, metric, and cutoff, with high specificity (range 0.95 to 0.99) and NPV (range 0.93 to 0.95). For variable type, it detected changes with high sensitivity (0.94 [0.91, 0.97]) and PPV (0.89 [0.82, 0.94]). For time point, it performed well in identifying both changes and no changes. For classifying types of changes (i.e. no change, more complete, less complete, change in definition), percent agreement ranged from 88.1% to 97.2% for individual elements (Figure 2; Supplementary Figure 3) and Gwet's AC values from 0.86 to 0.97 (Supplementary Figure 4).

The automated approach correctly identified 100% of outcomes with a priority change and 99.2% of outcomes without a priority change.

Supplementary Box 2 presents examples of GPT's rationale when determining the change status of outcomes.

Additional analyses

We did not find evidence of heterogeneity across prespecified subgroups. For identifying outcome changes, both percent agreement and Gwet's AC were generally high and consistent over time and across ACTs, NIH-funded trials, and other trials (Supplementary Table 4).

In *post hoc* analyses, the automated approach performed similarly across primary, secondary, and other outcomes (Supplementary Table 5). Differences among GPT-4o, o1, and o3 mini were small (Supplementary Figures 3 and 4). The "best two of three" approach performed slightly better than the individual models. Performance declined when full examples were omitted from the prompts.

Our approach demonstrated high accuracy in identifying types of change (Supplementary Table 6). All accuracy indices (e.g., PPV, NPV) suggested that our approach was as good or better at identifying changes in definitions compared with changes in completeness. For example, sensitivity was 0.99 (95% CI: 0.97–1.00) for detecting outcomes with at least one element that

changed in definition, and 0.95 (95% CI: 0.91–0.98) for detecting outcomes with changes in completeness only.

Post hoc sensitivity analyses suggest the results would not differ meaningfully if clustering of outcomes within trials had been accounted for when estimating the point estimates. We observed either no or weak associations between number of outcomes per trial and accuracy (most Spearman's $Rho < 0.3$; Supplementary Tables 7 and 8). Moreover, the average accuracy across outcomes and the average accuracy across clusters were not meaningfully different (Supplementary Tables 7 and 8).

Discussion

Our automated approach was very accurate at defining clinical trial outcomes, evaluating whether registered outcomes were defined completely, and identifying changes in outcome definitions and priorities. Agreement between the automated approach and consensus-answers by humans was comparable to agreement between human raters in this study and in previous studies.³⁸

Our approach could be used to improve registrations before trials are conducted and thus prevent concerns about reporting biases. Lack of understanding and resources might contribute to registrations that do not meet registries' quality control criteria, which can increase the time required for sponsors and registries to review records and to post them publicly³³ and lead to suboptimal registrations. Automated assessments of outcome definitions, including detailed explanations (Box 3), could help investigators, sponsors, regulators, and registries identify and fix errors and omissions before studies begin enrolling participants.

Our approach could also help investigators describe how definitions and priorities changed. For example, reporting guidelines recommend that investigators describe outcome changes,^{16,25,27} which are often subtle. Our approach could help editors, peer reviewers, and systematic reviewers efficiently identify these changes—information essential for assessing risk of outcome reporting bias.^{36,39}

Regulators might use this automated approach to flag changes that warrant review by humans, but it is not a substitute for judgment. For example, the U.S. Food and Drug Administration

defines changes that require prior approval before implementation, which include changes to primary outcomes.⁴⁰ Similarly, the European Medicines Agency might require approval for substantive changes that affect the reliability of trial data.⁴¹ Our approach could help users flag potentially significant changes, but it cannot judge whether changes are important or scientifically justified. For example, our approach does not consider the timing of changes, which might occur before much data have been collected and before studies have been unmasked, so even major changes to outcome definitions are not necessarily evidence of poor research practices or bias. We did not define or compare all components of the estimand framework (e.g., population, strategies for handling intercurrent events) because there are not structured fields to enter this information on ClinicalTrials.gov and other registries, and further research about estimands using statistical analysis plans and other documents might be informative. Improving current registration requirements and practices could reduce reporting biases. That is, laws and policies require that many trial registrations include some of the elements we assessed,^{42,43} but most prospective outcome definitions in our validation set were not complete. Although not our primary objective, this study is the largest and most comprehensive evaluation of outcome definitions to date, and our findings are consistent with previous studies that found most registered outcomes are not defined completely.^{24,30} Compared with one large study,¹⁹ we found a large proportion of outcome definitions changed between prospective registrations and posted results, perhaps because we defined each outcome with greater granularity.

Investigators might be more likely to define outcomes completely if journals and organizations such as the International Committee of Medical Journal Editors (ICMJE) were to require that outcomes be defined completely in prospective registrations. Further research might evaluate whether and how policies, including ICMJE's registration policy, have affected the completeness of outcome definitions on ClinicalTrials.gov.

Meta-researchers could use our approach to conduct large-scale analyses of structured information in registrations. Although protocols and SAPs are not available publicly for all registered trials, it is a limitation that we did not review those documents when they were available on ClinicalTrials.gov. Protocols and SAPs typically include more information about outcomes and analyses compared with registrations, but they might not be useful for comparing

results with *prospective* plans because protocols and SAPs are rarely available before trials begin. Instead, most protocols and SAPs are uploaded when results are posted,⁴⁴ consistent with applicable laws and policies.^{2,42} Nonetheless, additional research could examine automated approaches for identifying and extracting structured information about outcomes and analyses from protocols and SAPs and for comparing those documents with results on ClinicalTrials.gov and other reports, including journal articles. We expect there will be challenges associated with assessing protocols, SAPs, and journal articles. Before comparing different types of documents (e.g., a registration versus a protocol versus a journal article), improved methods to extract outcome information and relationships across multiple sections of documents might be necessary (e.g., outcomes defined partially in methods, results, tables, and figures). Of course, editors and regulators making high stakes decisions might be encouraged to use automated tools to support human review rather than to replace more precise evaluation.

It is a limitation that we did not perform a completely masked assessment of model output. We reviewed over 110,000 separate LLM evaluations, which would have been prohibitive to enter manually and to review before comparing human ratings with LLM ratings. For pragmatic reasons, two independent raters read text from ClinicalTrials.gov alongside one LLM's evaluation, and we then adjudicated any differences between human raters. This could have introduced automation bias if both evaluators subconsciously deferred to LLM-generated rationales. Risk of bias might be limited by the use of two independent raters. Moreover, the raters saw a single model's output rather than the final answer from the LLM-based approach, which was based on the consensus of three models.

We developed and validated our approach using hundreds of trials and thousands of outcomes and results. Because our sample was restricted to phase III or "not applicable" phase randomized trials registered on ClinicalTrials.gov, our results might not generalize to other study designs (e.g., single-arm trials). We were unable to assess whether the accuracy of our approach varied across other study characteristics, such as intervention type and masking status, and some characteristics were uncommon in our sample (e.g., drug-device combination products). Future research might investigate the prevalence of various outcome definitions and of changes, including sources of heterogeneity (e.g., intervention type, trial design).

Our automated approach did not use information from all registration fields because LLMs can become “lost in the middle” when inputs are long.⁴⁵ Including information from other fields—such as the trial summary and detailed numerical results from the results module—could help improve accuracy if future LLMs were less likely to be “lost in the middle” compared with current LLMs.

The LLMs we used could not process one trial with many outcomes. This problem affected 1% of eligible studies in our sample, and relatively few studies registered on ClinicalTrials.gov might encounter similar errors. Modifying our approach to evaluate outcomes in smaller groups might have allowed us to process trials with any number of outcomes, but a modified approach might have reduced consistency and increased the number of false positive detections of outcome changes. That is, we evaluated all outcomes from a single trial in a single model call because we observed during development that our approach most accurately compared outcomes between two registrations when all outcomes from both versions were processed together. At the time of our study, the lowest token limits were a maximum context of 128,000 tokens and a maximum output of 16,384 tokens (GPT-4o-2024-11-20). As LLM capabilities improve, and as input and output token limits increase, it might become possible to include more information from ClinicalTrials.gov and to process many outcomes within a single prompt.

It is also a limitation that we treated multiple time points as having equal priority. Investigators sometimes list a specific measurement once and enter multiple time points in a single field.^{20 21} In these cases, it might be unclear whether multiple time points will be analyzed together (e.g., repeated measures analysis) or separately. If they are analyzed separately, it might be unclear whether multiple time points have the same or different priority (e.g., one is considered primary and others secondary). Unless investigators provide the information elsewhere, it might be unclear which of multiple time points were used for study design (e.g., power analysis) and which were expected to be used for interpretation.

Future research could investigate heterogeneity across registries, LLMs, and health problems. Although ClinicalTrials.gov is the world’s largest register, the WHO and ICMJE recognize 18 primary registries for clinical trials.^{46,47} We might expect performance to be worse on other registries that are less structured, or structured differently, compared with ClinicalTrials.gov.

Some investigators and organizations might be hesitant to use proprietary LLMs, especially for trials that have not started recruiting or that are ongoing. Future studies might apply our approach using other models (e.g., models with publicly available parameters, such as LLaMA, DeepSeek). Although we expect the outcomes can be defined using the same elements regardless of the trial design, we did not encounter many more complex designs (e.g., factorial, stepped-wedge, MOST, SMART, and Bayesian adaptive trials that changed outcome definitions by design) which can be more difficult to register compared to parallel trials.⁴⁸ Larger studies would be needed to investigate possible sources of heterogeneity, including differences in accuracy across health problems, interventions, and trial designs.

In conclusion, our LLM-based approach is a scalable tool to define outcomes, assess outcome completeness, and detect outcome changes. It could help stakeholders improve trial registrations, identify outcome changes, and assess risk of bias.

Methods

Overview

As described in our prespecified protocol (<https://osf.io/2tyh3>),⁴⁹ we developed an automated approach using LLMs to define and compare all outcomes in the *prospective* and the *latest* registration of each ClinicalTrials.gov registration (Figure 4). We defined the *prospective registration* as the last version before each trial's actual or estimated start date. We defined the *latest registration* as the most recent version of each trial's registration at the time of our study (i.e., as of January 27, 2024). We used Python (3.13.2) scripts to scrape and format registry data, calculate textual similarity, and check whether two text strings were identical. We used LLMs for textual interpretation, summarization, and semantic comparison. We defined each element separately before comparison; this approach outperformed direct comparison, except for the time point, where direct comparison yielded better results. We compared outcome priority according to the fixed headings that investigators assigned on ClinicalTrials.gov. LLM outputs were benchmarked against human assessments of the original registry texts, which served as ground truth. This report follows the STARD 2015 guidelines for reporting diagnostic accuracy

studies.⁵⁰ Python scripts to execute the whole approach are freely available on GitHub (<https://github.com/xiying5/OutcomeCheck>).

Eligible trials

For prompt engineering and validation, we included only completed RCTs with outcomes registered prospectively (i.e., before the “Start Date” on ClinicalTrials.gov) and with results uploaded to ClinicalTrials.gov by the time of our study. We also required that trials had uploaded a protocol and a statistical analysis plan on ClinicalTrials.gov, but we did not use these additional documents for this study.

From the eligible trials, we selected trials randomly using stratified random sampling by study type, which included Applicable Clinical Trials (ACTs),^{42,51} NIH-funded trials, and other trials. We used 225 trials for training, which included 74 ACTs, 73 NIH-funded trials, and 78 other trials. We then randomly selected an additional 150 eligible trials for validation using the same stratified sampling approach. Random sampling was performed using Stata 16 (StataCorp LLC, College Station, TX, USA).

Rules for defining and comparing outcomes

Each outcome was defined using five of six elements that define an outcome—specific measurement, specific metric, variable type, cutoff, time point (Table 2).²⁹ We did not consider outcome domains because investigators are not prompted to enter domains on ClinicalTrials.gov and we could not infer them reliably.

Outcomes from the prospective and latest registration versions were matched one-to-one based on textual similarity (Figure 5). Outcomes without matches were categorized as “added outcomes” or “missing outcomes.” For matched outcomes, each of the five elements was compared then classified as no change, more complete, less complete, or change in definition (Table 2). An element was classified as *change in definition* if it was defined in both versions but the definitions differed substantively (e.g., a time point of 12 months versus 6 months). An element was classified as *more complete* if it was not defined in the prospective registration but was defined in the latest version (e.g., time point not specified versus specified as 12 months), and less complete in the opposite scenario. An element was classified as *no change* if the

element had the same definition in both versions or was not defined in either version. Clerical changes, such as correcting typos, rewording that did not alter the underlying meaning (e.g., week 1 versus day 7), or differences in explanatory information (e.g., descriptive information about thresholds for interpretation), were classified as *no change*.

Table 2. Rules to define outcomes and outcome changes

Element	Operational definition used in our automated approach	Change in definition*	Change in completeness†
Priority	The category assigned by the investigators on ClinicalTrials.gov: <i>primary</i> , <i>secondary</i> , <i>other</i> (for prospective registrations), <i>other pre-specified</i> (for latest registrations), and <i>post-hoc</i> (for latest registrations).	The priority category differed between versions (e.g., primary to secondary, secondary to other).	Not applicable No priority definitions were “Incomplete” for outcomes on ClinicalTrials.gov because outcomes must be entered under one of the required headings.
Specific measurement	The name of a scale or instrument used to assess an outcome, or the name of an outcome that can be observed directly. a. For a standardized scale or instrument used to assess a subjective characteristic: The name of the scale or instrument, or “Not reported” if unnamed. b. For a customized, unstandardized scale: The number and type of questions as well as the characteristics assessed, or “Not reported” if neither question number and type was reported. c. For a characteristic that	Scales differed between versions (e.g., differences in total/subscale, version, modifications). For custom questionnaires, a difference in (1) the number or type of questions, or (2) the characteristics being assessed (e.g., “3-question 5-point Likert scale assessing patient satisfaction with treatment” vs “1-question 4-point Likert scale assessing perceived benefit from treatment”).	More complete (1) Prospective version was “Not reported” or “Not valid” AND (2) Latest version was defined Less complete (1) Prospective version was defined AND (2) Latest version was “Not reported” or “Not valid”

	can be observed directly or assessed by common tests or devices considered interchangeable for research purposes: The name of the characteristic. d. For composite or multi-criteria outcomes: The name of each component/criterion, or “Not reported” if components/criteria are not listed. e. If multiple measures are mentioned and they do not form a single measurement: “Not valid”.	Characteristics that were observed directly or assessed by common tests or devices differed between versions (e.g., fasting blood glucose vs random blood glucose). Composite or multi-criteria outcomes: any component differed between versions, or one or more components were added or removed	
Specific metric	The type of value assigned to each individual participant for a specific measurement, which we categorized as: “Value at a time point,” “Change between two time points,” “Rate of change,” “Variability or average per person,” “Sum per person,” “Maximum/minimum per person,” “Events per person,” “Time-to-event,” or “Duration.” “Not reported” if a single metric could not be inferred. “Not valid” if multiple metrics were described and they did not form a single outcome.	The specific metric category differed between versions (e.g., “Change from baseline” vs “Rate of change”)	More complete (1) Prospective version was “Not reported” or “Not valid” AND (2) Latest version was defined Less complete (1) Prospective version was defined AND (2) Latest version was “Not reported” or “Not valid”

Cutoff	Information needed to define the value for each individual for some outcomes. When applicable, a cutoff might include: - The value being categorized; - The specific threshold or categories that will be compared; - Direction of interest (e.g., greater than, decrease) and operational signs (e.g., =, >, <, ≥, ≤). “Not applicable” if an outcome is not defined using a cutoff. “Not reported” if a cutoff is implied but not provided. “Not valid” if multiple non-mutually exclusive cutoffs are listed.	The value being categorized differed between versions (e.g., SF-36 Physical Functioning subscale score ≤50 vs SF-36 Social Functioning subscale score ≤50). The threshold differed between versions (e.g., “Fasting glucose > 100 mg/dL” vs “Fasting glucose > 126 mg/dL”). “Not applicable” in one version and reported in the other. “Not applicable” in one version and “Not valid” in the other.	More complete (1) Prospective version was “Not reported” or “Not valid” AND (2) Latest version was defined Less complete (1) Prospective version was defined AND (2) Latest version was “Not reported” or “Not valid”
Variable type	The method used to aggregate scores or values across individuals for an outcome. We described all outcomes as “continuous,” “categorical,” or “continuous and categorical”. “Not reported” if no explicit information is given.	The variable type chosen differed between versions (e.g., “continuous” vs “categorical”).	More complete (1) Prospective version as “Not reported” AND (2) Latest version was defined Less complete (1) Prospective version was defined AND (2) Latest version was “Not reported”

Time point§	The time(s) at which the values will be recorded to be used for analysis.	One or more time point(s) did not match (e.g., “12 and 18 weeks” vs “12 and 24 weeks”). A time point (excluding baseline) not included in one record is reported in the other record (e.g., “12 weeks” vs “12 and 18 weeks”).	More complete (1) Prospective version was “Not reported” AND (2) Latest version was defined Less complete (1) Prospective version was defined AND (2) Latest version was “Not reported”
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* An outcome could have changes in more than one element.

† For all elements, “defined” included anything other than “not reported” or “not valid”. For cutoff, “defined” also included “not applicable” (i.e., no cutoff was used to define the outcome).

§ Because investigators could enter multiple time points into a single field on ClinicalTrials.gov, we treated all time points with the same priority and did not consider whether all of them were used in a single analysis.

Data collection and preprocessing

For each trial, we downloaded fields listed in Table 3 from registrations using the ClinicalTrials.gov application programming interface (API).

Table 3. Input fields and number of prompts to define and compare each outcome element.

Element	Input fields*	Prompts to define element	Prompts to compare element
Specific measurement	For both versions: Outcome Measure Title, Outcome Measure Description, Outcome Measure Time Frame For latest version only: Row/Category Title	Outcomes in both versions were defined using the same three prompts: 1. Name Composition of Outcomes 2. Name “Specific measurement” for a Single Score or Event 3. Name “Specific measurement” for a Single Composite	One prompt that compares specific measurement of matched outcomes
Specific metric	For both versions: Outcome Measure Title,	Outcomes in both versions were defined using two different	None. Direct text string comparison done in

	Outcome Measure Description, Outcome Measure Time Frame For latest version only: Measure Type, Measure of Dispersion/Precision, Row/Category Title, Unit of Measure, Method for the Statistical Test of Hypothesis, Estimation Parameter	prompts: 1. Define Specific Metric for All Outcomes in the Prospective Version 2. Define Specific Metric for All Outcomes in the Latest Version	Python.
Cutoff	For both versions: Outcome Measure Title, Outcome Measure Description, Outcome Measure Time Frame For latest version only: Measure Type, Measure of Dispersion/Precision, Row/Category Title, Overall Number of Participants/Units Analyzed, Number of Participants/Units Analyzed, Unit of Measure, Method for the Statistical Test of Hypothesis, Estimation Parameter	Outcomes in both versions were defined using the same one prompt: 1. Define Cutoff for All Outcomes in the Prospective and Latest Versions	One prompt that compares cutoff of matched outcomes
Variable type	Same as specific metric	Outcomes in both versions were defined using two different prompts: 1. Define Variable Type for All Outcomes in the Prospective Version 2. Define Variable Type for All Outcomes in the Latest Version	None. Direct text string comparison done in Python.
Time point	Same as specific measurement	Compared directly without being first defined.	One prompt that compares time point of matched outcomes

*Field type: 1. Outcome Measure Type [Select one: Primary, Secondary, Other Pre-specified, Post-Hoc]; 2. Outcome Measure Title [Free text]; 3. Outcome Measure Description [Free text]; 4. Outcome Measure Time Frame [Free text]; 5. Measure Type [Select one: Count of Participants, Mean, Median, Least Squares Mean (LSM), Geometric Mean, Geometric LSM, Number, Count of Units]; 6. Measure of Dispersion/Precision [Select one: Not Applicable, Standard Deviation, Standard Error, Inter-Quartile Range, Full Range, _____ % Confidence Interval, Geometric Coefficient of Variation]; 7. Row/Category Title [Free text]; 8. Overall Number of Participants/Units Analyzed [Numeric value]; 9. Number of Participants/Units Analyzed [Numeric value]; Unit of Measure [Free text]; 10. Method for the Statistical Test of Hypothesis [Select one: Cox Proportional Hazard, Hazard Ratio (HR), Hazard Ratio (Log), Mean Difference (Final Values), Mean Difference (Net), Median Difference (Final Values), Median Difference (Net), Odds Ratio (OR), Odds Ratio (Log), Risk Difference (RD), Risk Ratio (RR), Risk Ratio (Log), Slope, Other (____)]; 11. Estimation Parameter [Select one: ANCOVA, ANOVA, Chi-Squared, Chi-Squared (Corrected), Cochran-Mantel-Haenszel, Fisher Exact, Kruskal-Wallis, Log Rank, Mantel Haenszel, McNemar, Mixed Models Analysis, Regression (Cox), Regression (Linear), Regression (Logistic), Sign Test, t-Test (1-Sided), t-Test (2-Sided), Wilcoxon (Mann-Whitney), Other (____)]; Based on #7, 8, 9, we derived a variable named Single Categorization, coded “Yes” if the sum of category-level participants equaled the overall total—indicating the result likely reflects a single outcome. This helped the LLM distinguish between single and multiple outcomes.

Prompt engineering and automated approach development

We iteratively developed prompts to instruct LLMs to define, match, and compare outcomes using 225 training trials (3,424 outcomes). In each iteration, we used a small batch of trials to test and refine the prompts; we revised prompts to resolve errors observed by human raters before moving to the next batch. After 225 trials, further adjustments no longer yielded notable performance improvements. We began development in November 2023 using GPT-4 and continued development using GPT-4o.

We found that performance improved when we divided the overall task into smaller subtasks, provided comprehensive instructions and examples to the LLMs, and asked LLMs to generate chain-of-thought rationales before outputting answers. The final approach comprised a series of few-shot prompts to first define each outcome element except time point. These prompts included examples with expert-written chain-of-thought reasoning⁵² and answers. Specifically, we used one prompt each to define specific metric, cutoff, and variable type (>10,000 words, excluding text inserted from each registration). Because specific measurements were more difficult to define, we used three prompts (>9,000 words, excluding text inserted from each registration). We also observed that performance in comparing outcomes between registration versions improved when outcomes from both versions were defined simultaneously by the LLM;

accordingly, outcomes in both versions of each trial registration were defined in a single model call. We used the same prompt to define the specific measurement and cutoff elements for both registration versions because the same type of information was available in both versions. When defining specific metric and variable type for outcomes in the latest registration versions, we added additional instructions to incorporate information from trial results to define these elements. Data fields used to define each element in each version are listed in Table 3.

Outcome matching by LLMs was guided by a few-shot prompt. To improve matching, we calculated Jaccard and Cosine similarity scores⁵³ between registration versions in Python, and we provided the Jaccard and Cosine scores to the LLMs alongside the original texts. The Jaccard score measures word overlap between two texts regardless of order, whereas the Cosine similarity accounts for both word frequency and relative position.

We developed zero-shot prompts to compare the defined *specific measurements* and *cutoffs*, and we used Python to compare *specific metric*, *variable type*, and *outcome priority*, which were defined by fixed categories. Prompt instructions to compare specific measurements, cutoffs, and time points were 695 words, 481 words, and 2,666 words long, respectively.

Illustrative excerpts from the prompts are shown in Box 3. Complete prompts are available on Open Science Framework (<https://osf.io/2tyh3>).

Box 3. Illustrative excerpts from prompts used to define, match, and compare outcomes

We developed eight prompts to define outcome elements, one prompt to match outcomes between registrations, and three prompts to compare the elements. Each prompt started instructions formatted as lists embedded with brief, real and invented examples, followed by long examples with expert-written chain-of-thought reasoning and answers. Each prompt ended with a description of the task and instructions to output both a rationale and a judgement. Outputs were formatted as structured JavaScript Object Notation (JSON) files.

Example excerpts are shown below.

....

Guidelines for Identifying and Defining "Metric"

-In clinical trials, outcomes are defined using multiple elements. "Metric" is one of those elements.

1. Definition of "Metric": The unit of measuring scores at the individual participant level. Possible answers: "End value", "Duration", "Variability or average per person", "Change from baseline", "Events per person", "Rate of change (repeated measures)", "Time-to-event", "Sum per person", "Maximum per person", or "Minimum per person".
-Sometimes "Not reported" or "Not valid".

2. Definitions of possible answers

B. "Duration": A continuous variable representing the amount of time per person

-The outcome is defined by the duration or length of time, or the proportion of time, during which an event or activity occurs or does not occur for each participant, and the event status can be observed for all participants during the study period.

-The answer is not "Duration" if the outcome assesses the number of occurrences of an event within a period of time.

Examples

- Length of hospital stay
- Duration of medication use
- Proportion of days with pain
- Number of drinking days per week

Examples

Question: "Outcome Measure: Incidence of at least 1 RBC transfusion or hemoglobin less than or equal to 8.0 g/dL from study day 1 to end of treatment period. Outcome Description: . Time Frame: study day 1 until end of efficacy treatment period."

What is the "Metric" of this outcome?

Rationale, Metric: The outcome measures occurrence of either RBC transfusion or hemoglobin less than or equal to 8.0 g/dL (low hemoglobin). "Incidence" means new cases, indicating the event can happen only once per person during the study period. "At least 1" also indicates the outcome is concerned with only the first occurrence of either event in each person, not the number of events. Thus, the answer is not "Events per person". There is not enough information to determine whether the outcome will be analyzed using the proportion of people with the event or using each participant's time-to-event (e.g., log-rank test).

Metric: Not reported

Task

To answer each question, think step by step. Define the “Metric” of each outcome listed in the input and provide the reasons for your answers as in the examples above. Do not use information from one outcome to derive the answer for another outcome. Treat each outcome independently. Respond as a clinician and scientist with medical knowledge, expertise in medical terminologies and knowledge of how to diagnose and measure the health conditions being assessed in this study, and training in epidemiology and biostatistics.

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**Validation**

We validated our approach using a separate random sample of 150 eligible trials (50 ACTs, 50 NIH-funded, and 50 Other). The overall sample size was chosen to estimate sensitivity and specificity with 5% precision at a 95% confidence level, assuming 75% sensitivity and specificity and prevalence of outcome changes of either 20% or 80%.<sup>49</sup>

We processed all outcomes from the prospective and latest trial registrations in the validation set in January 2025. For *GPT-4o-2024-11-20*, we set temperature at 0 and top\_p at 1. For *o1-2024-12-05* and *o3-mini-2025-01-31*, we set reasoning effort at medium. To minimize random errors, we selected the answer given by at least two of the models. When all three models provided different answers, we selected one of their responses randomly using Stata. The exception was the matching task, which was completed exclusively by GPT-4o because it performed near-perfectly and was cost-effective.

**Human rating**

Human raters (XY, KN, JPO, CV, SZ, AB, RQ, NJD, MJP) with expertise in medicine and epidemiology evaluated the trials independently and in duplicate. To facilitate their ratings, we imported information from the ClinicalTrials.gov registrations and LLM outputs into REDCap. Raters reviewed the original registry texts and model-generated answers side by side. Raters were instructed to verify all outcomes against the original registry content to identify any information that the LLMs might have missed or fabricated (“hallucinated”). All training trials were double-rated.

We evaluated 150 trials during the validation phase. We validated outcome *definitions* using 75 trials, and we validated outcome *changes* using 150 trials. That is, we randomly selected 75 trials for which two raters (XY, KN, JPO, CV, AB) evaluated both the definition of each element in the prospective registration and the definition of each element in the in the latest registration, and they also compared the matching and change status of each element between the two registrations. For 75 additional trials, two raters evaluated only whether outcomes were correctly matched and whether the change status of each element was correct. Although the validation trials were processed using three models, raters reviewed only the GPT-4o outputs and were not aware of the final model decision (i.e., the response agreed upon by at least two of the three models).

### Statistical analysis

We used descriptive statistics to summarize trial characteristics. We also assessed agreement between human raters and summarized the number of tokens and associated costs using current prices. Based on human ratings, we calculated the frequencies of registration completeness and change status for each element and for each outcome overall.

We compared the automated approach with human ratings for defining and comparing outcomes. For variables that could be assigned one of many categories (e.g., change in definition, more complete, less complete, no change), we calculated percent agreement (not corrected for chance) and Gwet's AC (corrected for chance).<sup>54,55</sup> For binary variables, we calculated sensitivity (recall), specificity, positive predictive value (precision, PPV), negative predictive value (NPV), c statistic, and F1 score (the harmonic mean of precision and recall).

The estimand of interest was average performance at the outcome level (using the statistics noted above).<sup>56</sup> Accordingly, we undertook analyses in which each outcome contributed equally to the point estimate.<sup>57,58</sup> That is, trials with many outcomes contributed more information than trials with few outcomes. In a sensitivity analysis, we examined a trial-level estimand in which each trial contributed equally to the point estimate. We also examined if the number of outcomes per trial was associated with the performance. Confidence intervals around these estimates were calculated via bootstrapping (5000 replications, percentile method). We allowed for clustering of outcomes within trials by resampling the entire trial.

We hypothesized that the automated approach would be more accurate for high-quality registrations compared with low-quality registrations. We therefore examined if measures of accuracy differed for newer versus older trials. We used July 1, 2018 as a cutoff because ClinicalTrials.gov increased quality control and began checking outcome definitions more consistently around this time. We also compared ACTs, NIH-funded trials, and Other trials because the first two groups might be reviewed most rigorously by sponsors and ClinicalTrials.gov.<sup>24</sup>

*Post hoc*, we evaluated the performance of LLMs in identifying outcome change by outcome priority. We compared the performance of individual LLMs in identifying outcome change status using percent agreement and Gwet's AC. Because including examples increased prompt length and associated costs, we also compared performance with and without detailed examples.

All analyses were performed with Stata 16.0 (StataCorp LLC, Texas, USA) and R version 4.2.1 (2022-06-23).

**Declaration statements:****(1) Data Availability**

Model outputs are available on OSF upon publication (<https://osf.io/2tyh3>).

**(2) Code Availability**

The code used to implement our approach are available on GitHub (<https://github.com/xiying5/OutcomeCheck>).

**(3) Acknowledgements**

This work was supported by the National Institutes of Health (R01LM014079) and the UNC Provost's AI Acceleration program cloud resource grant.

**(4) Author Contributions**

EMW conceived the study. EMW, HK, XY, CV, and AB obtained funding. XY and EMW wrote the prompts. XY and CV conducted prompting via the large language model API. HK and JDM provided technical support for prompt engineering and interaction with the models. XY downloaded the outcome data from ClinicalTrials.gov API, designed rating instruments in REDCap, and trained the raters. XY, KN, JPO, CV, SZ, AB, RQ, NJD, MJP, and EMW rated the model-generated results of the development and validation trials. XY analyzed the data. CV checked the prompting code. XY and EMW drafted the manuscript. EMW, XY, TL, JEM, MJP, IJS, JC, HK, NJD, NJB, and MO contributed to the interpretation of the data. All authors critically revised the manuscript through multiple iterations for intellectual content and approved the final version.

**(5) Competing Interests**

The authors declare no conflicts of interest.

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## Figure legends

### Figure 1. Accuracy of the automated approach for classifying completeness (not defined vs. defined) in prospective registrations (538 outcomes) of 75 RCTs\*

\*Overall, an outcome definition was considered *incomplete* if any of the elements (except time point) was *not defined*. For all elements, we considered *not reported* or *not valid* to be *not defined*. For cutoff, we considered *not applicable* to be defined (i.e., no cutoff was used). See Supplementary Box 1 for interpretations of sensitivity, specificity, PPV, NPV, F1 score, and c-statistic.

### Figure 2. Percent agreement between automated approach and human raters.\* (A) Agreement on each element in prospective registrations. (B) Agreement on each element in latest registrations. (C) Agreement between the automated approach and human raters in comparing each element between prospective and latest registrations. In panel C, agreement was calculated across the multiple categories of outcome change status: no change, more complete, less complete, or change in definition.

\*Time points were compared directly in a single prompt without being defined for each version first because this approach yielded better results. For this reason, time point is included in Panel C but not in Panels A and B. The dashed red line indicates a post hoc threshold of 90%, above which agreement is considered good.

### Figure 3. Sensitivity, specificity, predictive values, and F1 scores for binary classification of outcome change status (change versus no change) for 150 RCTs with 821 outcome pairs, overall and by each element

$F1\ score = (2 * [sensitivity * ppv] / [sensitivity + ppv])$ ;  $c\text{-statistic} = (\text{number of concordant pairs} + 0.5 * \text{number of ties}) / \text{number of all usable pairs}$ ; The vertical black line indicates a post hoc threshold of 0.9, above which performance is considered good. The total number of outcomes with any change shown in this figure is 689 (top right table), which differs from the 696 reported in Table 1. This discrepancy arises because the figure presents results only for outcomes that the LLM correctly matched (as we report matching and comparison accuracy separately), whereas Table 1 reports the outcomes that the human raters matched.

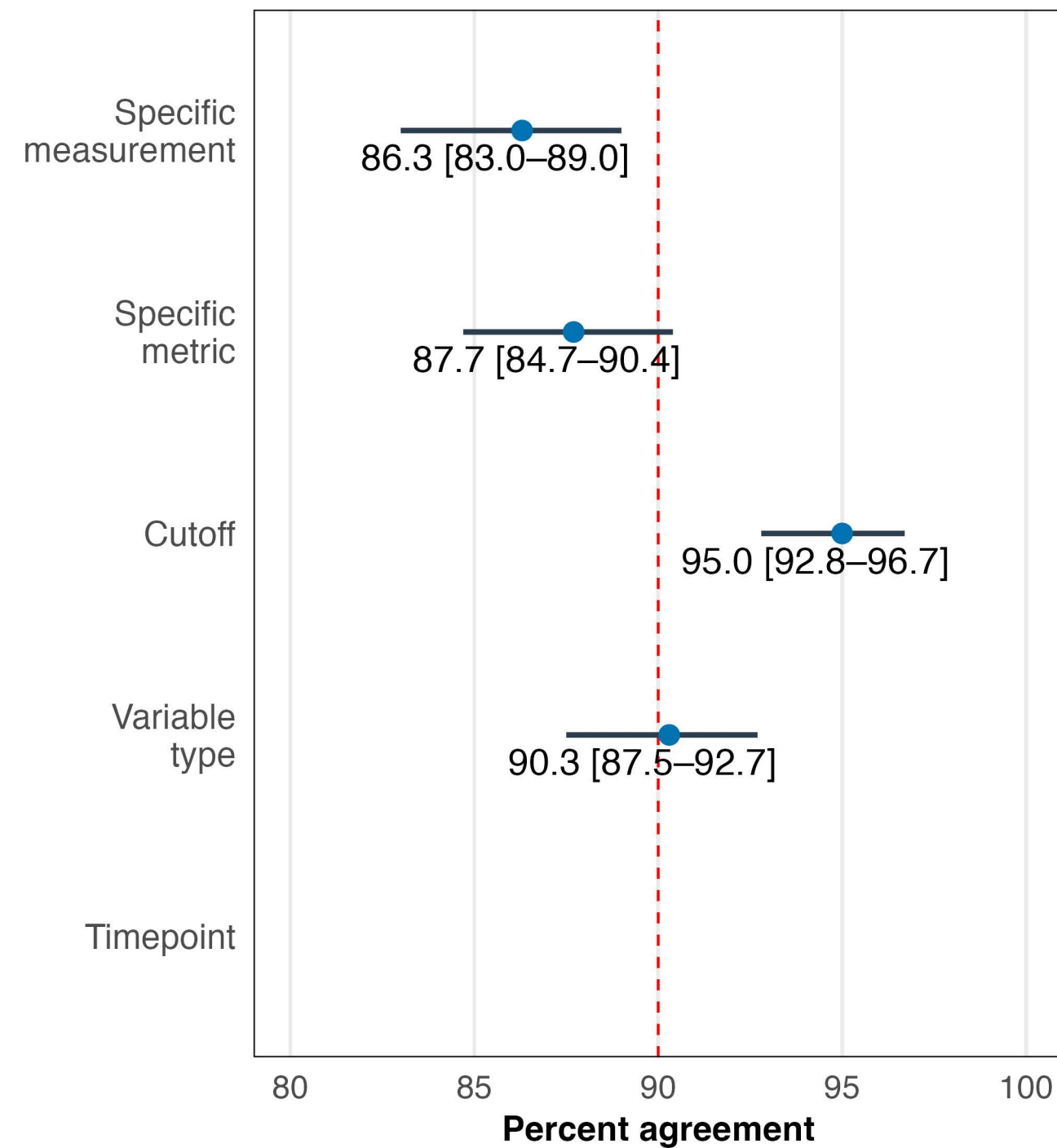
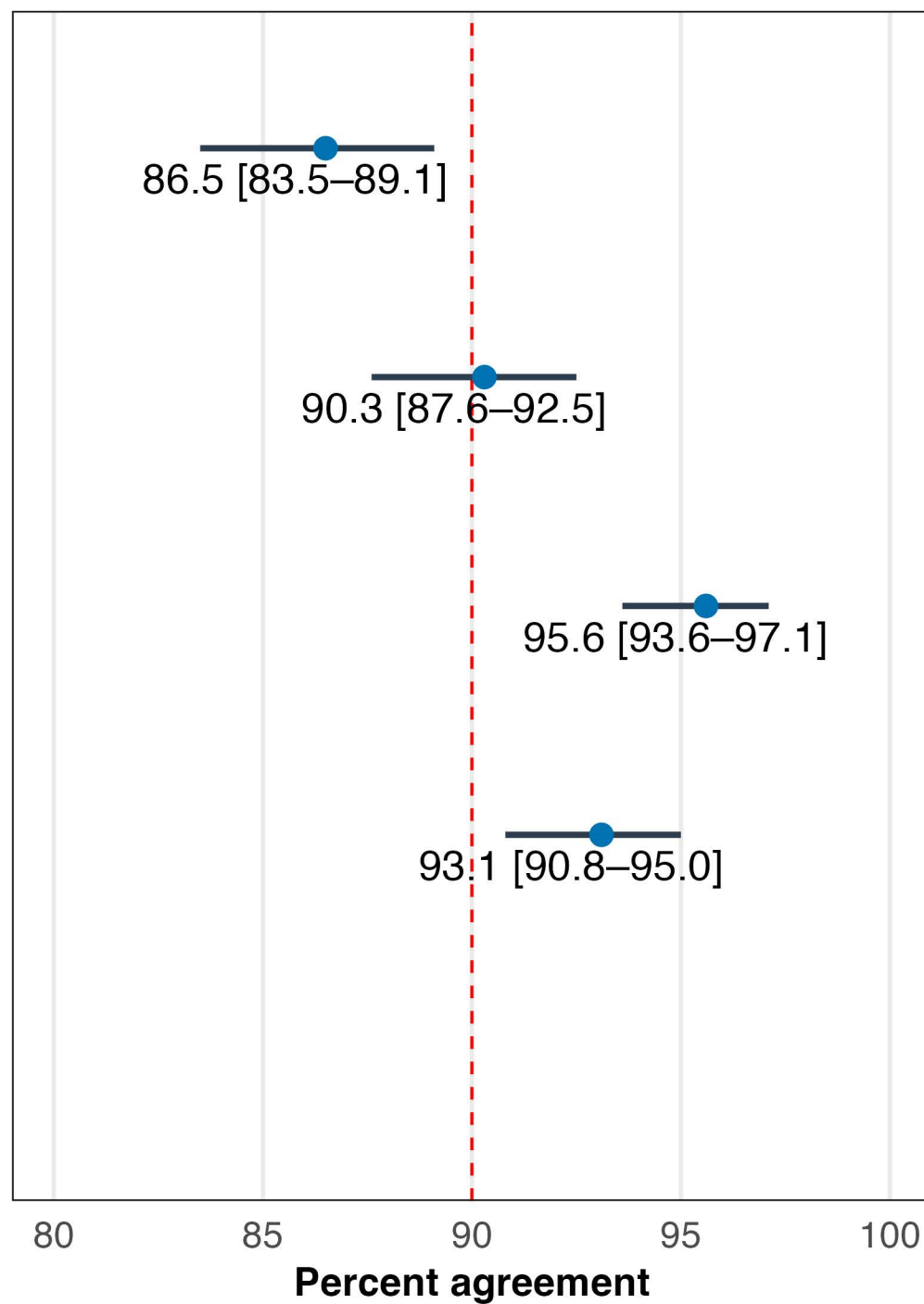
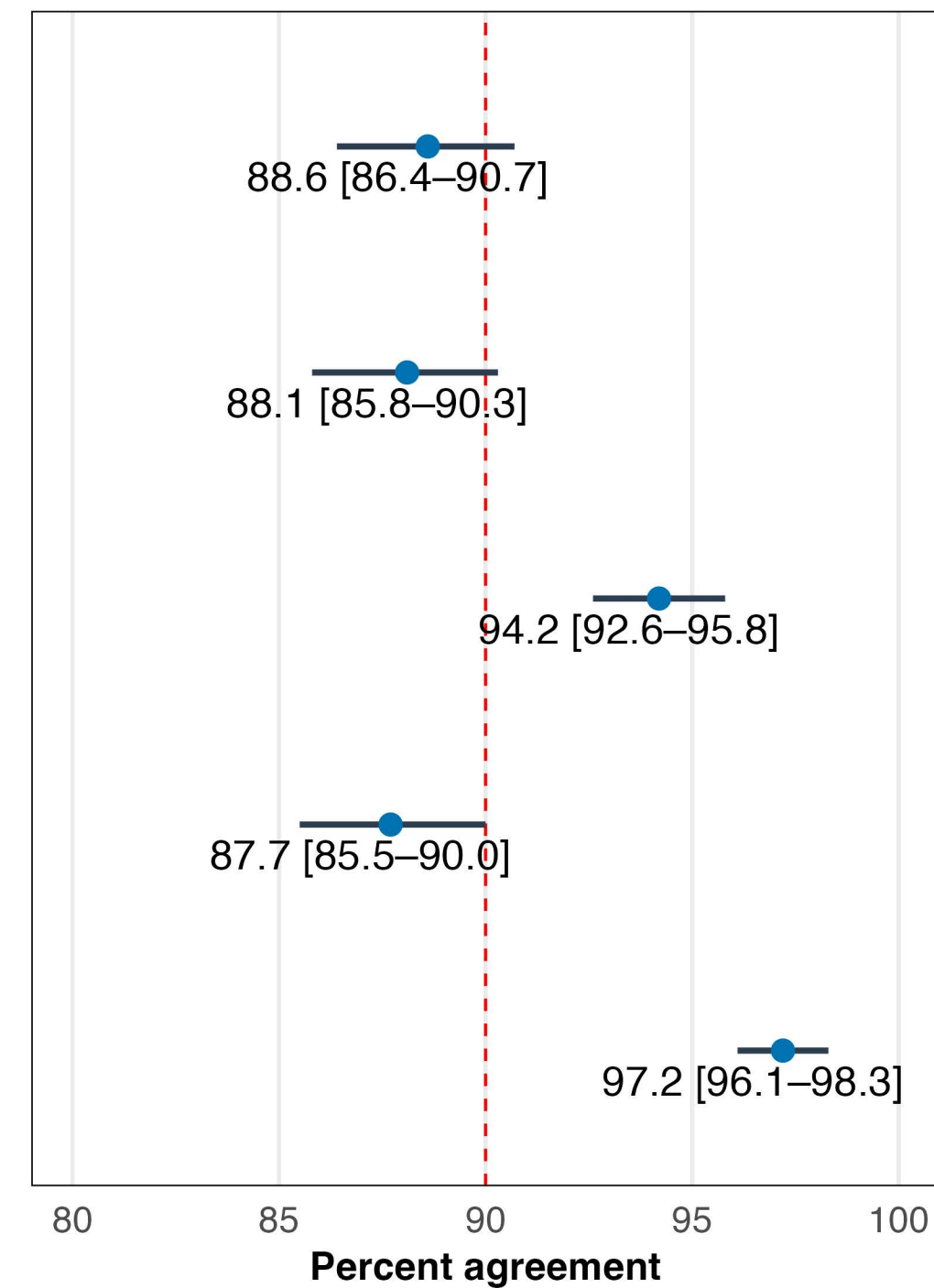
### Figure 4. Workflow of using LLMs to define and compare outcomes between the prospective and latest versions of trial registrations on ClinicalTrials.gov

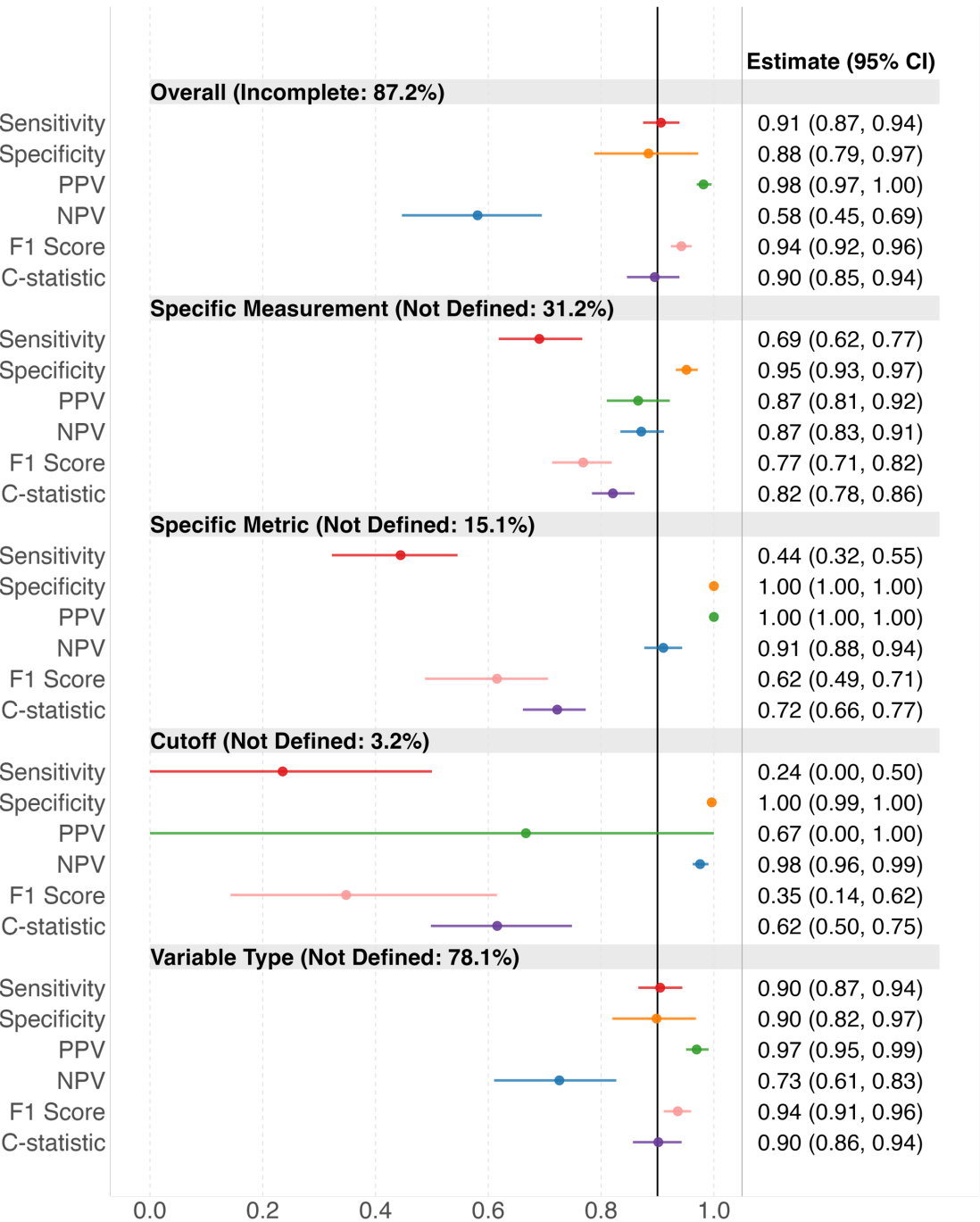
### Figure 5. Rules applied for outcome matching and in determining change status

From top to bottom: 1. One outcome added, one with no change—Outcomes were matched exactly when possible. 2. One outcome with change in definition, one outcome missing—When there was not an exact match, we matched each outcome in the prospective registration with the closest outcome in the latest registration. 3. One outcome with change in definition, one outcome added—When outcomes did not match exactly and there were multiple plausible matches, we matched each outcome in the prospective registration with one of the outcomes in the latest registration; For example, when a composite outcome in the prospective registration was split into several simple outcomes in the latest registration, one outcome in the latest version was

considered a correct match to the composite outcome and the remaining outcomes were categorized as *added*. 4. One outcome definition more complete, one outcome definition less complete—Matched outcomes had information about an element in one version that was not mentioned in the other version (i.e., the specific measurements “PHQ-9” and “VAS”).

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**A. Element definition (prospective)****B. Element definition (latest)****C. Element change (prospective vs latest)**



|             |                      |                  |
|-------------|----------------------|------------------|
|             | Actually not defined | Actually defined |
| Not defined | 426                  | 7                |
| Defined     | 43                   | 62               |

|             |                      |                  |
|-------------|----------------------|------------------|
|             | Actually not defined | Actually defined |
| Not defined | 114                  | 17               |
| Defined     | 54                   | 353              |

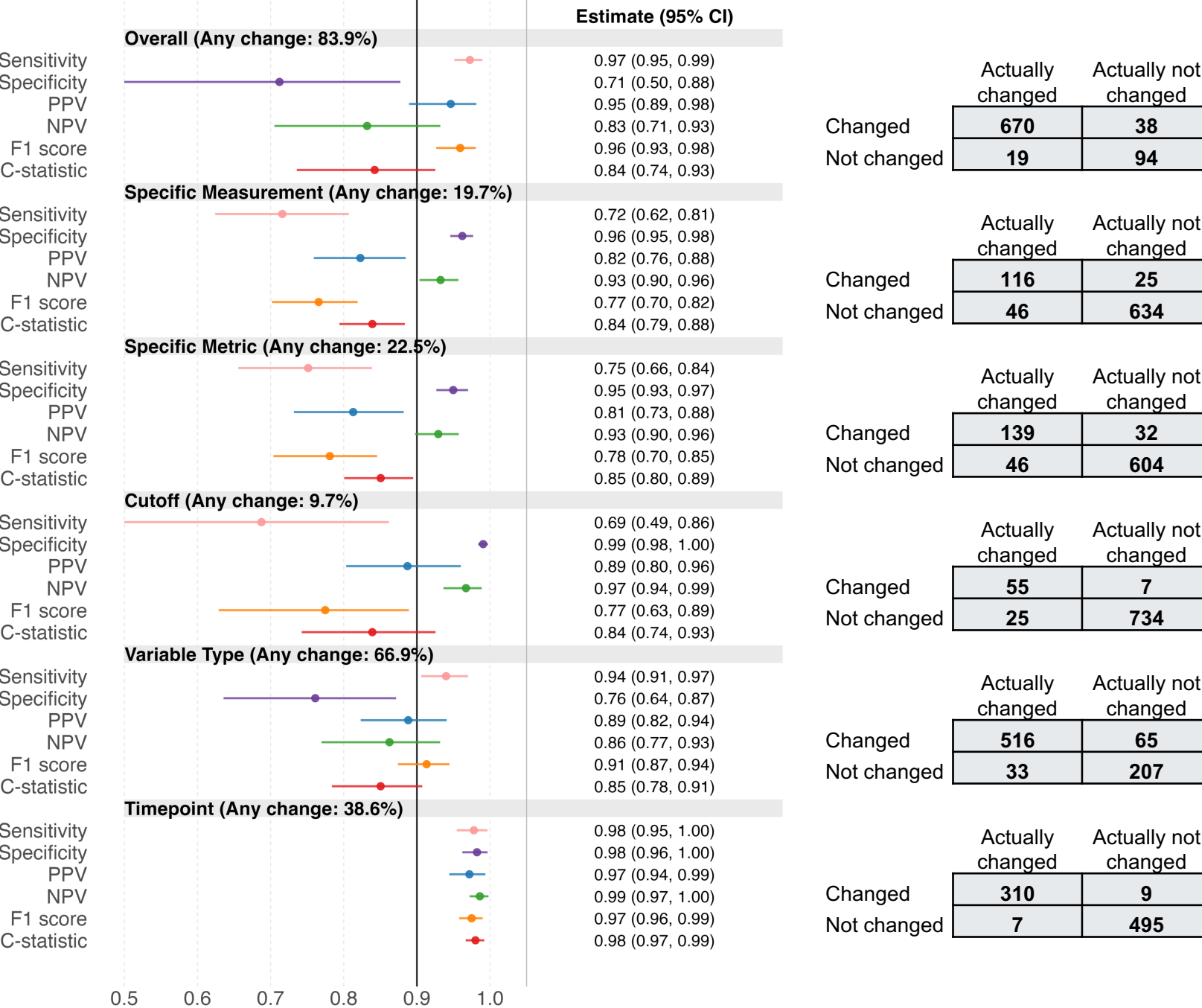
|             |                      |                  |
|-------------|----------------------|------------------|
|             | Actually not defined | Actually defined |
| Not defined | 34                   | 1                |
| Defined     | 47                   | 456              |

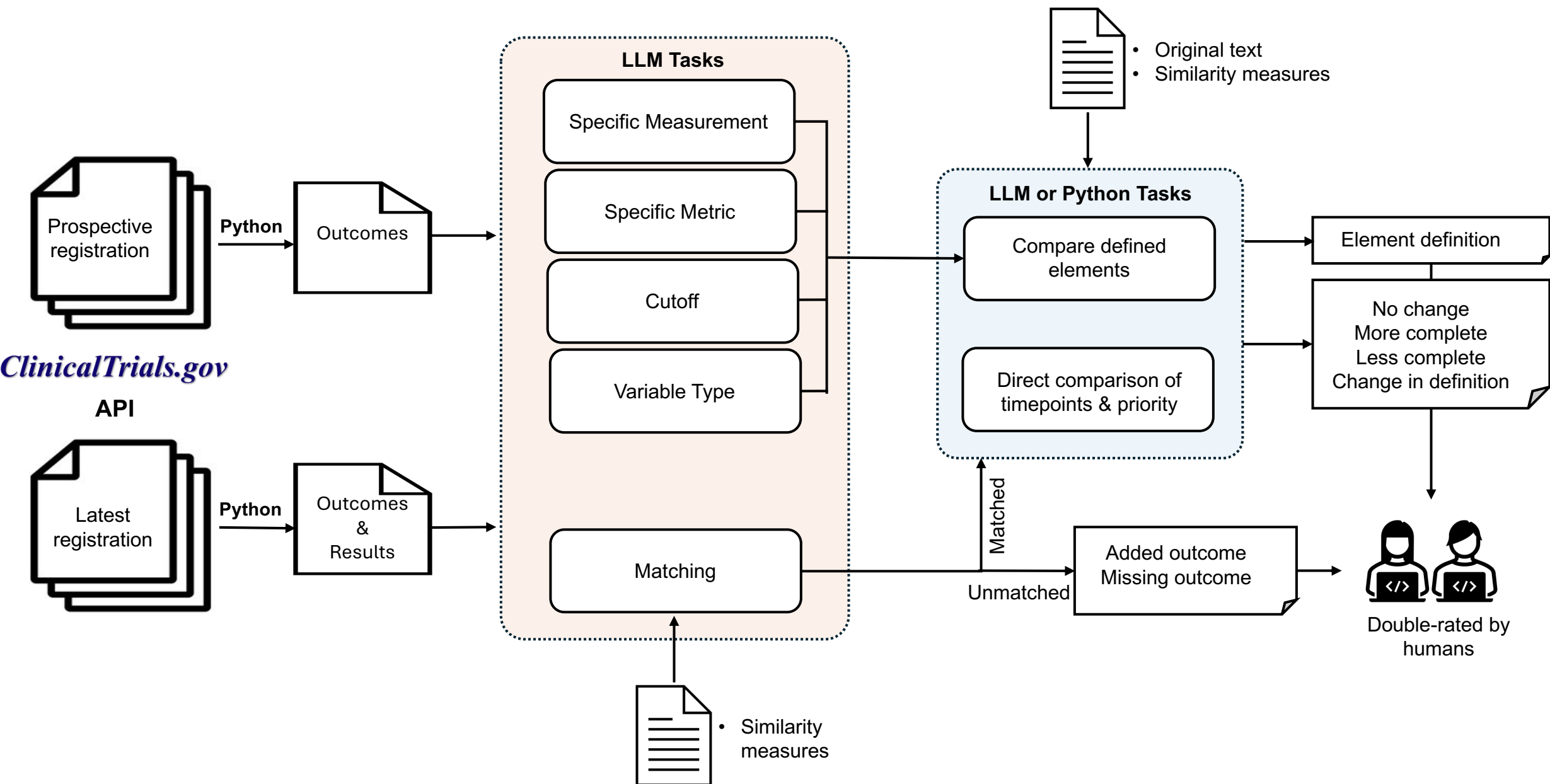
  

|             |                      |                  |
|-------------|----------------------|------------------|
|             | Actually not defined | Actually defined |
| Not defined | 4                    | 0                |
| Defined     | 13                   | 521              |

|             |                      |                  |
|-------------|----------------------|------------------|
|             | Actually not defined | Actually defined |
| Not defined | 380                  | 12               |
| Defined     | 40                   | 106              |







## Prospective registration

## Latest registration

1



All cause mortality

Cancer-specific mortality



Cancer-specific mortality

2

Cancer-specific mortality



All-cause mortality

Time to disease progression



3

Time to death  
or disease progression

Time to death



Time to disease progression

4

Depression



Depression (PHQ-9)

Pain intensity (VAS)



Pain intensity

## Change status



Matched



Added



Missing



No change



More complete



Less complete



Change in definition